THREATS TO CAUSAL INFERENCE. PART 2
MEASUREMENT ERROR AND INFORMATION BIAS

DOCTOR OF PUBLIC HEALTH PROGRAM
STAT 8311 - Causal inference in epidemiology
Lecture 6
Muhammad Adil Zainal Abidin

COURSE LEARNING OUTCOME

- Critically analyse theoretical concepts and assumptions underlying causal inference in epidemiology.
- Construct causal diagrams to represent causal relationships and pathways in epidemiological studies.
- Critically evaluate methodological threats to causal inference including confounding, **measurement error,** and missing data.
- Design statistical analyses to estimate causal effects in epidemiological data accounting for mediation, missing data, and third variables.

TOPIC LEARNING OUTCOME

By the end of this course, students will be able to:

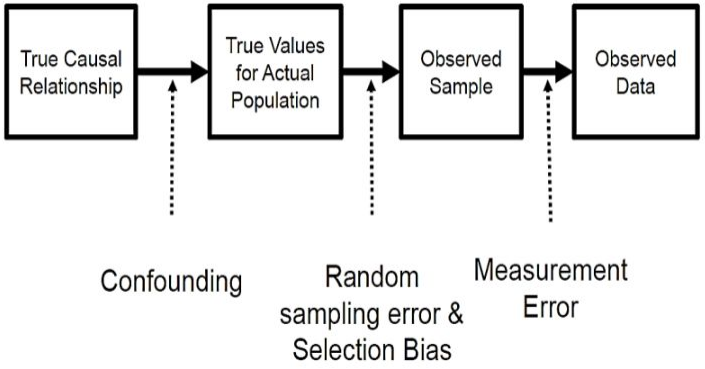
1. Define information bias and its sources
2. Identify type information bias and measure to control it.
3. Define measurement bias and its sources
4. Identify types of measurement bias and measure to control it.

BIAS

- Bias is any systematic error in an epidemiologic study that results in an incorrect estimate of the association between exposure and the health outcome.
- Bias is caused by systematic variation
- Bias occurs when an estimated association (risk ratio, rate ratio, odds ratio, difference in means, etc.) deviates from the true measure of association.

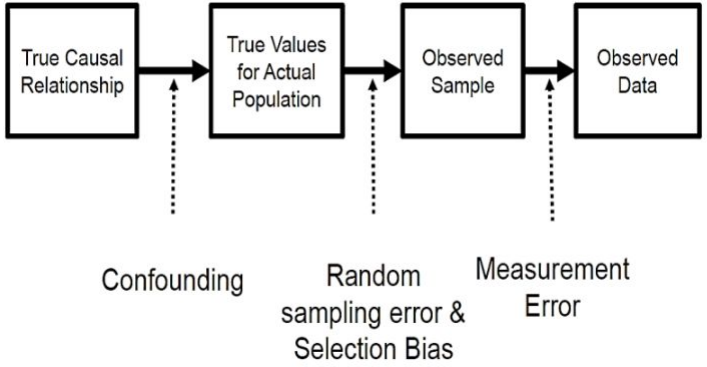
INFORMATION BIAS

- Information bias is a distortion in the measure of association caused by a lack of accurate measurements of key study variables.
- Information bias, also called measurement bias, arises when key study variables (exposure, health outcome, or confounders) are inaccurately measured or classified.
- Bias in the risk ratio, rate ratio, or odds ratio can be produced even if measured errors are equal between exposed and unexposed or between study participants that have or do not have the health outcome.



TYPE OF INFORMATION BIAS

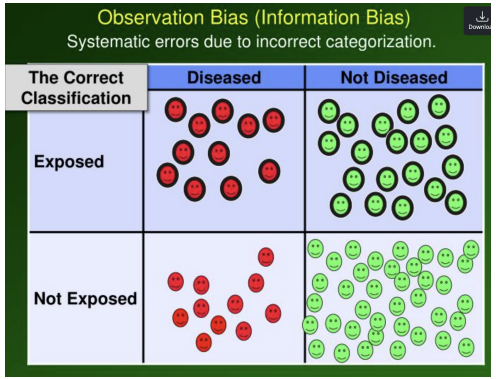
- Non differential misclassification bias
- Differential misclassification bias

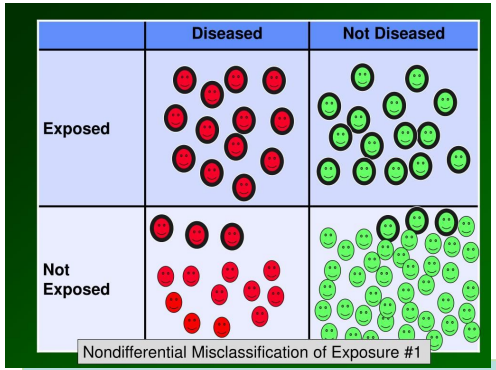


INFORMATION BIAS TYPE - NON-DIFFERENTIAL MISCLASSIFICATION

- Non-differential misclassification occurs if there is equal misclassification of exposure between subjects that have or do not have the health outcome or if there is equal misclassification of the health outcome between exposed and unexposed subjects.

- If exposure or the health outcome is dichotomous, then non-differential misclassification causes a bias of the risk ratio, rate ratio, or odds ratio towards the null.





Nondifferential Misclassification of Exposure #1

NON-DIFFERENTIAL MISCLASSIFICATION OF EXPOSURE STATUS

- Non-differential misclassification of exposure status in a case-control study occurs when exposure status is equally misclassified among cases and controls.
- Non-differential misclassification in a cohort study occurs when exposure status is equally misclassified among persons who develop and persons who do not develop the health outcome.
- Non-differential misclassification of health outcome status occurs in a case-control study when the health outcome status is equally misclassified among exposed and unexposed subjects.
- Non-differential misclassification of the health outcome status occurs in a cohort study when a study subject who develops the health outcome is equally misclassified among exposed and unexposed cohorts.

EFFECT OF NON-DIFFERENTIAL MISCLASSIFICATION OF EXPOSURE

- Non-differential misclassification biases the risk ratio, rate ratio, or odds ratio **towards the null** if the exposure classification is dichotomous, i.e., either exposed or unexposed.

- If exposure is classified into 3 or more categories, intermediate exposure groups may be biased away from the null, but the overall exposure-response trend will usually be biased towards the null.

EFFECT OF NON-DIFFERENTIAL MISCLASSIFICATION ON HEALTH OUTCOME

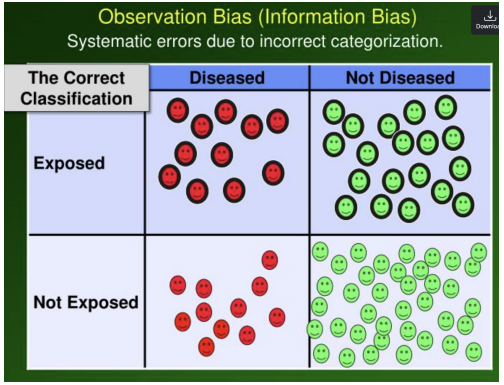
- In most cases, non-differential misclassification of the health outcome will produce bias toward the null, i.e. the risk ratio, rate ratio or odds ratio will be biased towards 1.0.
- If errors in detecting the presence of the health outcome are equal between exposed and unexposed subjects (i.e. sensitivity is less than 100%) but no errors are made in the classification of health outcome status (i.e. specificity is 100%), the risk ratio or rate ratio in a cohort study will not be biased, but the risk difference will be biased towards the null.

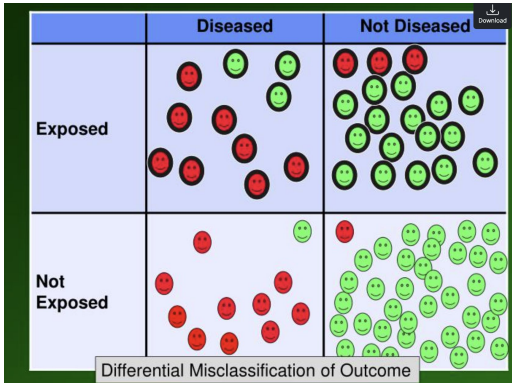
Effect of non-differential misclassification of health outcome status

- If no errors are made in detecting the presence of the health outcome (i.e. 100% sensitivity), but equal errors are made among exposed and unexposed in the classification of health outcome status (i.e. specificity less than 100%), the risk ratio, rate ratio, and risk difference (as applicable) will be biased towards the null.
- Combined errors in both sensitivity and specificity further increase the bias towards the null, but specificity errors produce larger biases overall.

INFORMATION BIAS TYPE - DIFFERENTIAL MISCLASSIFICATION

- Differential misclassification occurs when misclassification of exposure is not equal between subjects that have or do not have the health outcome, or when misclassification of the health outcome is not equal between exposed and unexposed subjects.
- Differential misclassification causes a bias in the risk ratio, rate ratio, or odds ratio either towards or away from the null, depending on the proportions of subjects misclassified.





Effect of differential misclassification of exposure or health outcome

- Differential misclassification of the exposure or health outcome can bias the risk ratio, rate ratio, or odds ratio either towards or away from the null.
- The direction of bias is towards the null if fewer cases are considered to be exposed or if fewer exposed are considered to have the health outcome.
- The direction of bias is away from the null if more cases are considered to be exposed or if more exposed are considered to have the health outcome.

Complications in predicting direction of misclassification bias

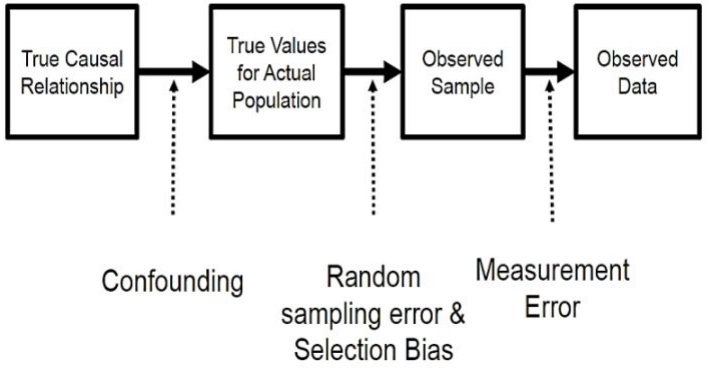
- Misclassification of confounders results in unpredictable direction of bias.
- Non-differential misclassification of a polychotomous exposure variable (3 or more categories) may result in bias away from null, though this is less likely than bias towards the null.
- Non-differential misclassification of a health outcome limited to a loss of sensitivity of detecting the health outcome without any loss in specificity does not bias toward null, whereas a loss of specificity always biases toward the null.

SOURCE OF INFORMATION BIAS

1. Recall bias

2. Interviewer bias – knows the status of the subjects before the interview process

3. Measurement bias



1. Source - Recall or reporting bias

- Recall or reporting bias is another form of information bias due to differences in accuracy of recall between cases and non-cases or of differential reporting of a health outcome between exposed and unexposed.
- Cases may have greater incentive, due to their health concerns, to recall past exposures.
- Exposed persons in a cohort study may be concerned about their exposure and may over-report or more accurately report the occurrence of symptoms or the health outcome.

Recall bias type

- Rumination bias - the presence of disease influence the perception of its causes

The mothers in the "case" group have spent months agonizing and ruminating over why this happened to their child, mentally replaying every single thing they ate, drank, or took while pregnant. Because they have been obsessively scanning their memories for answers, they vividly recall taking an over-the-counter pill once during week 6. The mothers in the control group had no reason to ruminate; they completely forgot they took the same pill.

- Exposure suspicion bias - the search for exposure to the putative cause when the doctor know the outcome status

When interviewing a healthy control, the doctor asks, "Did you ever work around metal dust?" The patient says "No, mostly just standard warehouse work," and the doctor moves on. However, when interviewing a patient diagnosed with the lung disease, the doctor's suspicion is triggered. They probe deeply: "Are you sure? Think about your warehouse job. Did you ever sweep up near a welding station? Did you ever smell burning metal? Take your time, think back 20 years."

- Participant expectation bias - in a trial if the patient knows what they receive may influence their answer

When asked, "Which ear did you hold your phone to 20 years ago, and for how many hours a day?", the patient's expectation forces their memory to comply. They recall holding the phone precisely to the right ear (where the tumor is) for 4 hours a day, whereas their actual historical use was much lower and split between both ears.

Reporting bias

Systematic distortion that arises from the selective disclosure or withholding of information by parties

- Obsequiousness bias - a type of response bias where research subjects or survey respondents systematically alter their answers to be sycophantic, overly agreeable, or tailored toward what they perceive the investigator or clinician wants to hear
- Family aggregation bias- refers to systematic errors or distortions that occur when measuring or interpreting the tendency of a disease, trait, or behavioral/cognitive pattern to cluster within families.
- Underreporting bias - occurs when respondents or data sources provide fewer instances, lower frequencies, or smaller amounts of an event, behavior, or condition than what actually happened

Controlling Recall or reporting bias

Solutions:

1. Add a case group unlikely to be related to exposure
2. Add measures of symptoms or health outcomes unlikely to be related to exposure
3. Use negative control exposure or outcome
4. Blinding the observer

2.Source - Interviewer / observer bias

Interviewer bias is a form of information bias due to:

1. lack of equal probing for exposure history between cases and controls (exposure suspicion bias); or
2. lack of equal measurement of health outcome status between exposed and unexposed (diagnostic suspicion bias)

Affect assessment in observational and intervention study

Controlling Interviewer bias

Solutions:

1. Blind data collectors regarding exposure or health outcome status

2. Develop well standardized data collection protocols, pilot, pretest

3.Train interviewers to obtain data in a standardized manner , TOT

4. Seek same information about exposure from two different sources, e.g. index such as data triangulation designed to detect and neutralize interviewer bias, such proxy respondent or other data record

Practice 1

	Have health out-come	Do not have health out-come
Exposed	200	210
Unex-posed	340	500

1) Researchers conduct a case-control study. The following table shows the true classification of exposure and the health outcome

(Note: these data are hypothetical and typically researchers would not know the true unbiased distribution of exposure and outcome).

a) Calculate the odds ratio
Now imagine that 50 people with the health outcome were misclassified as being unexposed and 20 people with the health outcome were misclassified as being exposed.

Practice 1

	Have health out-come	Do not have health outcome
Exposed		
Unex-posed		

b) Create the corrected 2x2 table
c) Calculate the odds ratio for the corrected table
d) In which direction was the misclassification bias?

Answer 1

	Have health out-come	Do not have health outcome
Exposed	230	210
Unex-posed	310	500

a) Calculate the odds ratio

Odds ratio = (200*500) / (340*210) =1.4

c) Calculate the odds ratio for the corrected table

Odds ratio = (230*500) / (310*210) =1.8

d) In which direction was the misclassification bias?

The bias was away from the null (the null value is 1.0).

Practice 2

2) Researchers conduct a case-control study of the association between the diet of young children and diagnosis of childhood cancer, by age 5 years. The researchers are worried about the potential for recall bias since parents are being asked to recall what their children generally ate, over a period of 5 years. Which of the following potential control groups would be most likely to reduce the likelihood of recall bias?

a) Parents of children with no known health problems
b) Parents of children with other known, diagnosed serious health problems (aside from childhood cancer)
c) Parents of children with other known, diagnosed minor health problems

Answer 2

2. Answer choice B is the best choice. Researchers should aim to have similar recall bias between both the case and control groups.

Parents of children who have childhood cancer, which is a serious health problem, are likely to be quite concerned about what may have contributed to the cancer.

Thus, if asked by researchers, these parents are likely to think very hard about what their child ate or did not eat in their first years of life.

Parents of children with other serious health problems (aside from cancer) are also likely to be quite concerned about any exposure that researchers ask about.

Therefore, these parents can be expected to recall exposures in a way that is more comparable with parents of children who have cancer.

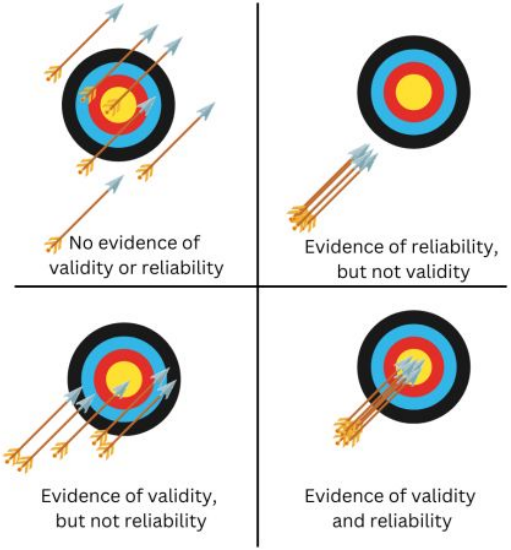
In contrast, parents of children who have no health problems or parents of children with only minor health problems are less likely to be concerned with carefully recalling any exposures.

3. MEASUREMENT BIAS

- As indicated above, errors in measuring exposure or disease can be an important source of bias in epidemiological studies In conducting studies, therefore, it is important to assess the quality of measurements.
- An ideal survey technique is valid (that is, it measures accurately what it purports to measure).
- Sometimes a reliable standard is available against which the validity of a survey method can be assessed.
- For example, a sphygmomanometer's validity can be measured by comparing its readings with intraarterial pressures, and the validity of a mammographic diagnosis of breast cancer can be tested (if the woman agrees) by biopsy.
- More often, however, there is no sure reference standard.
- The validity of a questionnaire for diagnosing angina cannot be fully known: clinical opinion varies among experts, and even coronary arteriograms may be normal in true cases or abnormal in symptomless people. The pathologist can describe changes at necropsy, but these may say little about the patient's symptoms or functional state.
- Measurements of disease in life are often incapable of full validation.

MEASUREMENT BIAS - VALIDITY AND RELIABILITY

- **Validity (Accuracy):** The extent to which a measurement or study actually measures what it intends to measure, and the degree to which causal inferences are true (absence of systematic error or bias).
- **Reliability (Precision):** The degree to which a measurement or study produces consistent, reproducible results under identical conditions (absence of random error).



MEASUREMENT BIAS -VALIDITY

Definition

In causal epidemiology, validity ensures that the observed association between an exposure and an outcome represents a true causal effect, rather than the result of flawed methodology, measurement error, or unmeasured variables.

Types of Validity

- **Internal Validity:** The extent to which study results accurately reflect the true causal relationship within the study sample. It is threatened by three main systematic errors:
    - Confounding: A third variable distorting the exposure-outcome relationship.
    - Selection Bias: Non-random selection or retention of study participants.
    - Information/Measurement Bias: Systematic errors in measuring exposure or disease (e.g., recall bias, misclassification).
- **External Validity (Generalizability):** The extent to which the study's causal findings can be appropriately applied to populations outside the study sample.
- **Measurement (Construct/Criterion) Validity:** How well a specific diagnostic test or metric represents the true biological state (e.g., does a survey truly capture "stress"?).

How to Conduct Statistical Analysis for Validity

- **Types**
  - **Criterion Validity -** When your measurement gives a "Yes/No" or "Disease/No Disease" result, its validity is evaluated against a Gold Standard (the definitive truth, like a biopsy or PCR test) using a 2x2 contingency table.
  - **Criterion Validity:** Continuous Measurements using ROC and Youden Index
  - **Construct Validity**: Surveys and Scales when measuring latent variables that have no physical "gold standard" (e.g., Quality of Life, Dietary Patterns, Patient Anxiety), you cannot calculate sensitivity or specificity. Instead, validity is established through statistical correlation and structural modeling.
  - **Convergent Validity:** Testing if your new survey correlates highly with existing, established surveys that measure the same concept.
  - **Discriminant (Divergent) Validity:** Testing to ensure your survey does not correlate with surveys measuring entirely different concepts.

How to Conduct Statistical Analysis for Validity

- **Evaluating Measurement Validity:**
  - **Criterion Diagnostic Accuracy**: Calculating Sensitivity, Specificity, Positive Predictive Value (PPV), and Negative Predictive Value (NPV) against a "Gold Standard."
  - **Criterion - ROC Curves**: Plotting Receiver Operating Characteristic curves and calculating the Area Under the Curve (AUC) to evaluate the discriminatory power of a continuous variable.
  - **Factor Analysis (EFA/CFA):** A complex dimensionality-reduction technique used to prove that the individual questions on your survey accurately map to the underlying theoretical concepts (factors) you claim they measure.
  - **Convergent Validity -**Test used: Pearson's r or Spearman's rho correlation coefficients. You want a high correlation (e.g., $r > 0.7$).
  - **Discriminant (Divergent) Validity:** Test used: Pearson's r. You want a low correlation (e.g., $r < 0.3$).

DIAGNOSTIC ACCURACY -ANALYSING VALIDITY

- When a survey technique or test is used to dichotomised subjects (for example, as cases or non-cases, exposed or not exposed) its validity is analysed by classifying subjects as positive or negative, firstly by the survey method and secondly according to the standard reference test.
- The findings can then be expressed in a contingency table as shown below.

New Test	Subjects with the disease	Subjects without the disease	Total
Positive	a (TP)	b (FP)	a +b (TP + FP)
Negative	c (FN)	d (TN)	c +d(FN + TN)
Total	a +c (TP + FN)	b +d(FP + TN)	a+ b+ c+ d (TP+TN+FP+FN)

Table 2: Assessment of new diagnostic test accuracy in relation to the gold stan-

DIAGNOSTIC ACCURACY - ANALYSING VALIDITY

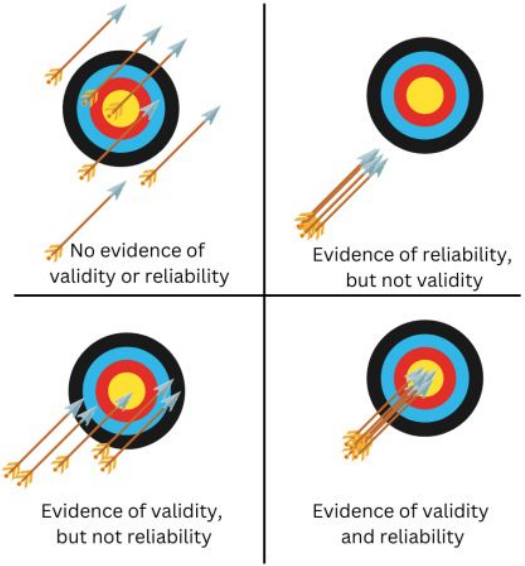
From this table four important statistics can be derived:

- Sensitivity – A sensitive test detects a high proportion of the true cases, and this quality is measured here by a/a + c.
- Specificity- A specific test has few false positives, and this quality is measured by d/b + d.
- Systematic error – For epidemiological rates it is particularly important for the test to give the right total count of cases. This is measured by the ratio of the total numbers positive to the survey and the reference tests, or (a + b)/(a + c).
- Predictive value-This is the proportion of positive test results that are truly positive. It is important in screening.
- It should be noted that both systematic error and predictive value depend on the relative frequency of true positives and true negatives in the study sample (that is, on the prevalence of the disease or exposure that is being measured).

MEASUREMENT BIAS - RELIABILITY

Definition

Reliability is the stability and reproducibility of epidemiological data. High reliability means that if you measure the same variable or conduct the same study multiple times, you will get highly similar results. It minimizes "noise" (random error), which is necessary (but not sufficient) for establishing causal validity.



MEASUREMENT BIAS - RELIABILITY

Types of Reliability

- **Test-Retest Reliability:** Consistency of a measurement when administered to the same subject at different times.
- **Inter-rater (Inter-observer) Reliability:** Consistency of a measurement when assessed by two or more different researchers or clinicians.
- **Intra-rater (Intra-observer) Reliability:** Consistency of a measurement when assessed by the *same* researcher on multiple occasions.
- **Internal Consistency:** The degree to which different items on a single epidemiological survey or scale measure the same underlying construct.

MEASUREMENT BIAS - HOW TO CONDUCT STATISTICAL ANALYSIS FOR RELIABILITY

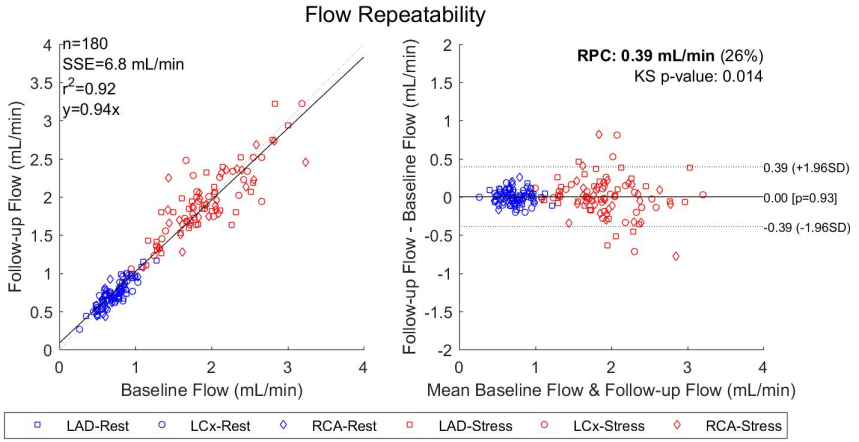
For Categorical/Nominal Data (e.g., Disease Yes/No):

- Cohen's Kappa (kappa): Measures inter-rater agreement for two raters, accounting for the possibility of agreement occurring by chance.
- Fleiss' Kappa: Used when assessing agreement among three or more raters.

MEASUREMENT BIAS - HOW TO CONDUCT STATISTICAL ANALYSIS FOR RELIABILITY

For Continuous Data (e.g., Blood Pressure, BMI):

- Intraclass Correlation Coefficient (ICC): The primary metric for continuous data, assessing how strongly units in the same group resemble each other (used for inter-rater and test-retest).
- Bland-Altman Plots: A visual statistical method plotting the difference between two measurements against their average, used to identify systematic bias between raters or machines.
- Coefficient of Variation (CV): Measures the standard deviation as a percentage of the mean to show variability.



MEASUREMENT BIAS - HOW TO CONDUCT STATISTICAL ANALYSIS FOR RELIABILITY

For Survey/Scale Instruments (e.g., Quality of Life scores):

- Cronbach's Alpha : Calculates average correlations among survey items to ensure they consistently measure the same latent trait (values > 0.70 generally indicate acceptable internal consistency).

THANK YOU

